RDBMS Object Creation Manual

March 9, 2018

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Introduction 2

Overview of the Objects Creation Procedure 3

Requirements 3

Suggestions 3

Scripts 4

Script Application 4

Creation Steps by RDBMS 6

- SQL Server 6
- Oracle 9
- PostgreSQL 11

SAP Hana 14

Conclusion 15

Glossary 16

Appendix A - Scripts in the JEE Ear Archive 17

Appendix B - Liquibase 18

Notes and Copyright 19

Introduction

This manual covers the steps to install the necessary application schemas/databases on compatible versions of the RDBMS instance(s) which will host the application data. This is one of two required steps to the complete the installation of the CPM application, the other being the deployment of the CPM application on the JEE application server instance(s). The procedures described here are also applicable to the creation a new schema/database for an already installed application.

During the installation of a new CPM application, the installation of the schema(s)/database objects part of the installation procedure should be coordinated with the application server procedure as the deployment of the application on the JEE AS requires the schemas/databases.

For compatible versions of each RDBMS and Editions, please see the CCH Tagetik Compatibility Matrix or the CCH Tagetik Hardware and Software Architecture Overview available on the Tagetik community or from Tagetik for more details.

Finally, if the reader has any questions or comments, please do not hesitate to contact CCH Tagetik.

Overview of the Objects Creation Procedure

The installation of the schemas/databases for the application consists in creating the application objects and configuring the RDBMS to allow access to these objects from the JEE application server (AS).

The objects need to be created either manually, or via scripts. These object all include the tables, views, functions, stored procedures, and indexes necessary for the application schema(s)/database objects. Additional objects may be necessary depending on the RDBMS such as logins.

The CPM application utilizes three distinct types of schema(s)/database(s) which are described below. Only the creation steps of the first two are covered in this manual.

- **Repository**: This required schema/database holds the application, user, and some project level configuration as well as the audit log. A project utilizes one repository type schema/database per environment, i.e. one repository for a production environment, one repository for a test environment, etc.
- **Business Data**: This required schema/database type holds all the business data of the project. Generally, the CPM application generally utilizes one schema/data per environment though, others can be utilized as well.
- **Secondary**: The schema/database type holds data which is not based on the CCH Tagetik application data models. Creation of this type of schema\database is not covered in this manual.

Requirements

This section describes some of the general requirements both technical and practical for creating the schema(s)/database objects on the RDBMS instance(s). Where necessary more specific details will be provided in the section regarding each RDBMS.

- **Administrative Privileges:** The reader will require an administrator or DBA user with privileges to install the required objects and to check and configure the RDBMS instance(s). For example, “sysadmin” “system”, “sa”, or “postgres” depending on the RDBMS utilized.
- **SQL Scripts:** For the schema(s)/database objects creation, CCH Tagetik provides SQL scripts. The following “Scripts” provides this information.
- **Tool for Launching the SQL Scripts**: It will be necessary to have a tool for applying the SQL scripts preferably, which allows logging during the script application.
- **Storage Available**: Each schema\database requires the appropriate amount of storage which depends on the project characteristics. For standard RDBMS (i.e. with the exception of SAP Hana), an absolute minimum of 10 GB per application schema\database is recommended, but more storage may need to be provisioned due to project characteristics and age. If a resource estimate is available from CCH Tagetik, this value may be greater. Adequate storage space for the RDBMS system files should also be provisioned.

Suggestions

This section presents some general suggestions and best practices which may be convenient to consider and/or implement during the installation of the application schema(s)\database(s).

Backups: Since all the application configuration and data is stored in the schema(s)\database(s), these should be backed up according to your IT policies. This topic is also covered in more detail in the CCH Tagetik documentation regarding maintenance procedures.

DB Maintenance: We recommend frequent rebuilding of indexes and recalculation of statistics or the equivalent procedures according to the RDBMS as these activities may have a positive impact on the operation of the application. The default settings regarding index and statistics maintenance of MS SQL Server and Oracle may not be adequate and these activities should be configured.

Performance: With regards to performance, the RDBMS instance configuration and its host platform should keep in mind that all the application data is stored in the databases and that the application can be very I/O intensive in bursts. Therefore, attention should be paid to the configuration of the RDBMS instance utilization of storage, placement of the database files, memory, and also to the characteristics of the storage solution. Consideration should also be paid to the processors, and we recommend selecting CPUs which favor single thread performance and high clock speeds.

CCH Tagetik recommends that its databases be hosted on a dedicated instance and on a physical host for performance considerations if at all possible, but neither is it required for the correct functioning of the application. The quantity of server resources such as processing power, memory will impact a project's performance and it is highly recommended that an estimate of suggested resources be requested from Tagetik or from your partner.

Scripts

This section details where the necessary SQL creation scripts can found. The scripts are available via three channels. The two primary channels are described below. The scripts are also present within the JEE ear archive file for CPM versions 5.1.2 and later is described in Appendix A.

1. **Installation DVD:** New customers should receive an installation DVD which contains all the files necessary for installing the CPM application. Regarding the schema(s)/database(s) installation, the necessary scripts are found in the sub-directories of **`...\database\<RDBMS>\`**,
   a. **Business Data:** …**`\database\<RDBMS>\Db_data\`**, i.e. for Oracle, …**`\database\oracle\Db_data\`**
   b. **Repository:** …**`\database\<RDBMS>\Db_repository`**, i.e. for PostgreSQL, …**`\database\postgres\Db_repository\`**
2. **Community Portal**: All customers receive access to the CCH Tagetik Community portal, and the schema/database scripts can be found in a zip file in the Installation and Upgrade CCH Tagetik Section. Here the reader would need to choose the destination version and download the zip file from the "Database scripts for new installations - ***<Destination Version>***" entry.
   For example, if the destination version is 5.3 (SP2), then the reader would download the zip file found at the entry "Database scripts for new installations - 5.3 (SP2)".
   After downloading the file, the reader would decompress the file and access the directory which corresponds to their RDBMS, i.e. select the mssql directory to access the SQL scripts to be used with SQL Server.
   Regarding the schema(s)/database objects installation, the necessary scripts are found in the sub-directories of **`...\<RDBMS>\`**,
   a. **Business Data:** …**`\<RDBMS>\application\`**, i.e. for Oracle, …**`\oracle\application\`**
   b. **Repository:** …**`\<RDBMS>\repo\`**, i.e. for PostgreSQL, …**`\database\postgres\repo\`**

Script Application

CCH Tagetik provides the scripts necessary for creating the database objects inside the schema(s)\database(s) by the channels described in the previous section. The application order and details of these SQL scripts depends on the version of the CPM application.

CPM Versions 5.1.1 and All Later Versions

For all versions of the CPM application from 5.1.1, the scripts should be applied in the following order. The scripts and their names for each RDBMS are valid for the scripts downloaded directly from the CCH Tagetik Community portal in zip files (see the previous section) and those provided by with the installation DVD. Scripts from the ear file have different names.

1. **Customization Script**: **`customization.sql`** - The script should be run on each application database, both the repository and each business database. The script creates the CONFIG_NAME_DB table and inserts some information necessary both for the application server to connect to the schema\database and some information to permit the usage of Liquibase (see the Appendix B for more information). See the section for each RDBMS for how the values should be specified.

Field Name	Value	Required
COD_DATABASE	Repository: [!] required Business: any string with no spaces	Required for all RDBMS.
DESC_DATABASE	Optional description of the database.	Optional
TABSPACE_DATA	Tablespace name used for data.	Required for Oracle
TABSPACE_INDEX	Tablespace name used for indexes.	Required for Oracle
TABSPACE_TEMP	Tablespace name to be used for temp.	Required for Oracle
DBSCHEMA	schema value	Required for MS SQL Server and PostgreSQL

2. **Creation Script**: **`tcpm_create.sql`** (business data) or **`rep_create.sql`** (repository) - These scripts should be applied on the appropriate schema(s)\database(s).

CPM Versions Earlier than 5.1.2

For version of the application earlier than 5.1.2, the following order should be followed.

1. **Creation Script**: **`tcpm_create.sql`** (business data) or **`rep_create.sql`** (repository) - These scripts should be applied on the schema(s)\database(s).
2. **Customization Script**: **`customization.sql`** - The script should be run on each application business database. The script creates the CONFIG_NAME_DB table and inserts some information necessary for the application server to connect to the schema\database. The common value for all RDBMS is the **`COD_DATABASE`** and **`DESC_DATABASE.`** The **`COD_DATABASE`** field needs to be set to the logical name of the schema\database.
3. **Patch Script**:

   `rep_<CPM version>_patch <number>_from_ear_<ear patch number>.sql` (repository) or

   `tcpm_<CPM version>_patch_<number>_from_ear_<ear patch number>.sql` (business data)– Optionally, it may be necessary to apply a patch script.

Creation Steps by RDBMS

The following sections present the steps for installing the objects necessary for the application by each RDBMS with which the CPM application is compatible.

SQL Server

This section covers the installation and configuration of the necessary database objects, i.e. databases, logins, on Microsoft SQL Server. These configurations and steps are valid for all compatible versions of MS SQL Server.

SQL Server Instance Requirements:

- **TCP/IP Enabled:** The SQL Server instance should have TCP/IP configured and enabled. The SQL Server instance will communicate with the application server via TCP/IP.
- **Case Insensitive Collation:** Each schema(s)\database(s) for each RDBMS instance should share the same case insensitive collation. For example, **`Latin1_General_CI_AS`**, could be used. It is also essential that the collation be consistent with the system databases such as tempdb.

Database Creation

Create each database using the standard method for creating a database in SQL Server. The following table presents the values suggested for the configuration of the database during the creation steps. The names listed below follow Tagetik conventions, but any names are possible.

Note: The values are merely examples, and these must be configured according to the specific environment/project.

Initial Parameters for Repository & Business Database(s)		
Parameter	**Value**	**Notes**
Database Name	**TGK_REPOS or TGK_DATA_001**	These values are solely examples.
Owner	<default>	Any value is possible.
Initial Size (mdf)	1024	Values are in MB. Using powers of 2 is suggested.
Initial Size (log)	512	Values are in MB. Using powers of 2 is suggested.
Autogrowth	By 128 MB, unrestricted growth	We suggest using unrestricted growth by 128 MB up to 1024MB, i.e. 128, 256, 512, 1024. The value should be chosen to favor performance of the RDBMS when the schema \ database requires additional storage space.
Path		This value concerns the location of the files on disk and should be configured by the customer's RDBMS storage conventions.
Recovery Model	Simple	We recommend the recovery model be set to **`Simple`** to avoid the saturation of storage resources as the application is very I/O intensive and may produce large log files. Other values are possible, but require appropriate configuration to avoid saturation of the available storage resources.

Login Creation

The application databases must be mapped to a SQL Server login so that they can be accessed by the application server. This login is usually dedicated to the application databases, i.e. not used for other purposes. Either SQL Server authentication or Windows Authentication can used.

Parameter	Value	Notes
Login name	tagetikcpm	Another example could be “tagetik”, any value could be used.
SQL Server authentication	Enabled	If disabled, i.e. it is necessary to use Windows Authentication. In either case, the username and password must be referred to the person configuring the application server. If using Windows authentication, the domain must also be referred.
Enforce password policy	Disabled	If necessary to enable this, attention must be paid to password expiration date, as AS data source configurations will need to be updated with password.
Default database	<repository dbname>	By convention, set default database to the repository database.
Default language		There are no specific value recommended.
Server Role	Public	Public is the only role required.
Map	Enable for <dbname>	Select the Tagetik databases.
User	tagetik login name	Value must be same as the name of Tagetik login.
Default Schema	dbo	
Database role membership for: <dbname>	“db_owner” & “public”	Required role memberships

Application of Scripts

After completing the steps of creating the database and the login described in this section, the next step is to apply the SQL scripts from CCH Tagetik described above. The scripts should be applied in the order according to the version of the application using the administrative, i.e. “sa” user. The spool or log files should be examined for errors and any errors should be referred to Tagetik.

Customization for the Databases

Customize the **`customization.sql`** script by inserting the name of the database in the fields **`COD_DATABASE`**, replacing the string “COD_DATABASE_TO_CHANGE” and in the **`DESC_DATABASE`** field, replacing the string “DESC_DATABASE_TO_CHANGE”. For application versions 5.1.1 and later, the following values are necessary to set for MS SQL Server.

Field Name	Value	Required
COD_DATABASE	Repository: [!] required Business: Name of the database	Required for all RDBMS.

DESC_DATABASE	Optional description of the database.	Optional
DBSCHEMA	dbo	Required for MS SQL Server.

Oracle

This section covers the installation and configuration of the necessary objects, i.e. schemas, users, on Oracle. These configurations and steps are valid for all compatible versions of Oracle.

Required Resources

The schemas to be created are based on the following resources:

- tablespace creation;
- user creation;

We generally recommend that two tablespaces be dedicated to the application to contain data and the indexes. Moreover, according to the application behaviour the temporary tablespace has to be provisioned appropriately for the expected usage. The following table shows example values, and the size characteristics must be configured according to the specific environment/project.

Suggested Tablespace Characteristics	
Tablespace Name	**Size**
TGK_DATA	20000 MB (minimum suggested)
TGK_INDEX	10000 MB (minimum suggested)
TEMP	10000 MB (minimum suggested)

The users to be created are related to the repository and to the business data schemas. Each user has to be granted with the following rights,

Roles	Connect, Resource
System Privileges	Alter User, Create Table, Create View, Create Procedure, Unlimited Tablespace, Select Any Dictionary

Object Creation

The objects belonging to the schemas are created by executing suitable scripts delivered as part of the installation package. Note: the scripts require the user to specify the tablespaces to be used.

- Data Tablespace
- Indexes Tablespace
- Default Tablespace (ex. the data tablespace)

Application of Scripts

After completing the steps of creating the schemas and the user described in this section, the next step is to apply the SQL scripts from CCH Tagetik described above. The scripts should be applied in the order according to the version of the application using the administrative, i.e. “system” or “sysadmin” user. The spool or log files should be examined for errors and any errors should be referred to Tagetik.

Customization for the Schemas

Customize the **`customization.sql`** script by inserting the username in the fields **`COD_DATABASE`**, replacing the string "COD_DATABASE_TO_CHANGE" and in the **`DESC_DATABASE`** field, replacing the string "DESC_DATABASE_TO_CHANGE". For application versions 5.1.1 and later, the following values are necessary to set for Oracle.

Field Name	Value	Required
COD_DATABASE	Repository: [!] required Business: name of schema	Required for all RDBMS.
DESC_DATABASE	Optional description of database.	Optional
TABSPACE_DATA	Tablespace name used for data.	Required for Oracle
TABSPACE_INDEX	Tablespace name used for indexes.	Required for Oracle
TABSPACE_TEMP	Tablespace name used for temp.	Required for Oracle

PostgreSQL

This section covers the installation and configuration of the necessary database objects, i.e. logins, databases, and extensions on PostgreSQL. These instructions were written with PostgreSQL version 9.4.5 installed but should be applicable to recent versions of PostgreSQL. It is assumed that an instance of PostgreSQL has already been installed on the server.

N.B. Compatibility with PostgreSQL was introduced with Tagetik 5.0.1 and earlier versions of Tagetik are not compatible with PostgreSQL.

In this manual, Tagetik will utilize the convention of creating a PostgreSQL database for each of the required Tagetik databases, i.e. the repository and one or more business database(s), rather than one postgreSQL database and two (or more) schema. The motivation is this approach provides increased security in the broadest number of potential customer contexts. Other configurations may be possible.

Requirements

The following items are required.

- **Admin Access**: The reader should have access to the "postgres" user which can create users, run scripts.
- **Network Connection**: The server hosting postgreSQL should be network-connected so as it is able to accept TCP connections (port numbers according to the installed software and the design of the specific installation). The default installation of PostgreSQL permits access only from localhost.
- **Extension uuid-ossp**: Application databases require that the PostgreSQL extension, **uuid-ossp** be installed on each database. See the section concerning this installation below.
- **Implicit Casts & Round**: Application business databases require an additional script to be applied to each one. This script, **`Cast.sql`**, enables a few limited implicits casts between data types and modifies the round function for the application databases. This script is available from the Community Portal and from the installation DVD. It is not found inside the JEE ear archive file. See the last sub-section of PostgreSQL Section for more details.

Required Resources

The databases to be created are based on the following resources:

- Login creation
- Extension
- Tablespace

The application does not have specific recommendations concerning the creation of the tablespace. It is usually recommended that two tablespaces be dedicated to the application to contain data and the indexes respectively. PostgreSQL default tablespaces can be utilized.

Login Creation

The application requires a login in PostgreSQL in order to allow the application server to access the data in the application databases. By convention, "tagetikcpm" or "tagetik" is used but any string may be used. This login is given ownership of the application databases. This step should proceed the creation of the databases. Here is an example command:

```
CREATE ROLE "tagetikcpm" LOGIN PASSWORD 'tagetikcpm' VALID UNTIL
'infinity';
```

Database Creation

The application requires a minimum of two databases. By convention, the database which contains the application configuration information, the user information, the audit logs, archived data, etc. is referred to as the "repository" and the logical name for this DB generally includes the string "*REPOSITORY*" or "*REPOS*"

for clarity. The second required database (and any additional ones) contains the business data of the project. Finally, all of the CPM application databases must be associated with a PostgreSQL login to allow the JEE application server to access these databases.

In psql or another tool, run the following commands to create the repository and the business databases and assign ownership to the login created earlier.

1.
```
CREATE DATABASE "TGK_REPOS"
WITH OWNER="tagetikcpm"
      ENCODING='UTF8'
      TEMPLATE=template0
            LC_COLLATE='C'
            LC_CTYPE='C';
\c TGK_REPOS
```
2.
```
CREATE DATABASE "TGK_DATA_001"
WITH OWNER="tagetikcpm"
      ENCODING='UTF8'
      TEMPLATE=template0
            LC_COLLATE='C'
            LC_CTYPE='C';
\c TGK_DATA_001
```

Repeat step 2 for any additional business databases.

In pgAdmin, it also possible to use the graphical interface by right-clicking on any item in the Object Browser on the left-hand side of pgAdmin.

Extension

Tagetik requires an extension library to available and accessible in the instance of postgreSQL. In Linux versions such as Ubuntu and Debian, this library can be installed with the command apt-get:

1. **apt-cache search uuid-ossp** (used to identify the name of the library)
2. **apt-get install postgresql-contrib-9.4**

Once the library is installed, the extension needs to be created in each Tagetik database in the postgreSQL instance.

1. **psql**
2. **\connect TCPM_DEMODEV_001**
3. **CREATE EXTENSION "uuid-ossp";**

In Windows, the uuid-ossp extension files are already provided with the installation files, check in the **...\share\extension** directory of the postgreSQL instance. When using the plugin PSQL Console or the Query tool, while connected to the desired database, only step 3 , i.e. **CREATE EXTENSION "uuid-ossp"** is necessary.

Application of Scripts

After completing the steps of creating the database and the other objects described above, the next step is to apply the SQL scripts from CCH Tagetik described above. The scripts should be applied in the order according to the version of the application using the application user defined earlier. The spool or log files should be examined for errors and any errors should be checked. Some examples regarding the application of the scripts are presented here.

- `psql -U tagetikcpm -W  -h localhost -d TGK_DATA_001 -f /home/tcpm_create.sql > stout.log 2>&1` (linux)

or

- **`psql -U tagetikcpm -W  -h localhost -d TGK_DATA_001 -f /home/tcpm_create.sql > stout.log 2>&1`** (Windows)

Customization for the Databases

Customize the **`customization.sql`** script by inserting the name of the database in the fields **`COD_DATABASE`**, replacing the string "COD_DATABASE_TO_CHANGE" and in the **`DESC_DATABASE`** field, replacing the string "DESC_DATABASE_TO_CHANGE". For application versions 5.1.1 and later, the following values are necessary to set for MS SQL Server.

Field Name	Value	Required
COD_DATABASE	Repository: [!] required Business: Name of the database	Required for all RDBMS.
DESC_DATABASE	Optional description of the database.	Optional
DBSCHEMA	dbo	Required for PostgreSQL.

Cast.sql

The **`Cast.sql`** script should be applied to each business database with the administrative user, i.e. "postgres" user. Due to the nature of the script, it will be necessary to re-apply this script after each restore of a backup, and after upgrades.

SAP Hana

This is a short how-to for create database users and schema for Tagetik application.
Use SAP HANA Studio to create users and run database creation script
HANA Studio must be configured to run script with a big number of statement:
- Open Windows-> Peferences menù
- filter on "SQL"
- set "Maximum Numbers of Characters for Multiple Statement Execution" to 2147483647

Create users in Sap HANA Studio
A dedicated HANA user should be created for each database.
Create Tagetik Application user(s) with the default granted rights.
Create Tagetik Repository user with the default granted rights.
Run database creation script in Sap HANA Studio
Open an SQL Window and connect to application database
- open, customize and run customization.sql
- open and run tcpm_create.sql
- open and run last db patch (ex. patch_Av000.x_from_ear_n)

Open an SQL Window and connect to repository database
- open, customize and run customization.sql
- open and run rep_create.sql
- open and run last db patch (ex. patch_Rv000.y_from_ear_n)

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Conclusion

Upon completion of the above procedures, the reader should perform the following tests and provide the information listed in the Connection subsection to the person installing the application server.

Connection Information

The following information needs to be recorded and made available to the person installing and configuring the application server(s). The following information is necessary for the configuration of the data sources which allow the application server to communicate with the Tagetik databases.

- All information necessary to connect to the RDBMS instance hosting the schema(s)\database(s) including the port.
- The user or login name with the password.

Glossary

The following table provides a short glossary for the terminology which are used in this manual.

AS	Application Server. This middleware software should be understood to mean JEE compliant Java application servers such as WildFly, JBoss EAP, IBM WebSphere, or Oracle WebLogic.
DB	Database and can also be used to mean schema
RDBMS	Relational Database Management System, i.e. MS SQL Server, Oracle, PostgreSQL, et..
JEE	Java Platform Enterprise Edition
DBAs	Database Administrator

Appendix A - Scripts in the JEE Ear Archive

For versions 5.1.2, and later, it is also possible to extract the creation SQL scripts from the JEE ear archive file. The JEE ear archive file is available in the following locations:

- **Installation DVD**: in the …**`\deployment\ear\`** directory.
- **Community Portal**: in the same section of the portal as the database script zip file.

Using a zip decompressor, i.e. Winzip (Windows), decompress the CPM application ear archive file and access the **`sqlscript`** directory. The creation scripts are found in the sub-directories are as follows,

- **Business Data**: …**`\sqlscript\<RDBMS>\application\`**, for example, for SAP Hana, the scripts would be found in …**`\sqlscript\hana\application\`** sub-directory.
- **Repository**: …**`\sqlscript\<RDBMS>\repo\`**, for example , for Postgres, the scripts would be found in …**`\sqlscript\postgres\repo\`** sub-directory.

There are differences in naming conventions and contents of the scripts. Application is in order.

1. **Customization Script**: **`customization.sql`** - the name and usage is the same.
2. **Creation Scripts**: **`repo_install_<RDBMS>.sql`** (repository) or **`application_install_<RDBMS>.sql`** (business data) - These correspond to the other creation scripts. However, these scripts do not contain all the same commands as the scripts available in the installation DVD or downloaded from the Community. The scripts with “update” must also be run.
3. Update Scripts: **`repo_update_<RDBMS>.sql`** (repository) or **`application_update_<RDBMS>.sql`** (business data) - These scripts are required and must be applied after the creation scripts.

Appendix B - Liquibase

With the addition of the Liquibase engine to the application in version 5.1.2, it is possible to set the application server to automatically create the schema(s)\database objects during the JEE ear deployment step for any RDBMS which is compatible with the application. This is generally done with the following JAVA system property set to true if all schemas\databases must be created. This is done at the AS level which configures the Java JVM.

`-Dit.grupposervizi.easy.liquibase.should.run=true`

With this property set to true, the embedded liquibase engine will apply the creation scripts to each of the connected (via the configured data sources, i.e. the repository and the primary application DBs, though, not the secondary databases) to the installation. This system property is the principal means for activating the automatic upgrade or installation utilizing the built-in motor. Two other optional system properties are also available which allow the administrator to provide more precise instructions to the liquibase motor.

`it.grupposervizi.easy.liquibase.update.repo=false`
`it.grupposervizi.easy.liquibase.update.app=false`

These system properties can be set to true or false. Setting these properties to false will disable the automatic update for the repository (repo) database or the business (app) databases. By default, they are set to true and are active when the primary system property is set to true.

Attention: Please check with the DBA if any before setting this parameter if it was not set during the installation procedure. Please note, that the only logging of the script application will occur in the AS instance server log and/or the tagetik server log, i.e. tagetik.log.
N.B. PostgreSQL requires the application of the **`Cast.sql`** script which cannot be run via the liquibase engine.

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